Royal Devon University Healthcare NHS Foundation Trust

Standard Operating Procedure
Guidance to Set-up Arterial Blood Pressure Monitoring

Northern Neonatal Services

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Please *specify* standard/criterion numbers and tick ✓ other boxes as appropriate

Monitoring Information		**Strategic Directions – Key Milestones**	
Patient Experience		Maintain Operational Service Delivery	
Assurance Framework		Integrated Community Pathways	
Monitor/Finance/Performance		Develop Acute Services	
CQC Fundamental Standards Regulations No:		Delivery of Care Closer to Home	
		Infection Control	
Other *(please specify)*:			
Note: This document has been assessed for any equality, diversity or human rights implications			

Controlled document

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Full History		Status: Draft	
Version	Date	Author (Title not name)	Reason
1.0	*13/04/2025*	*SSN Alicia Rogers*	*New document, to guide Health Professionals in the setup of Arterial Blood Pressure Monitoring (Northern Only)*

Associated Trust Policies/ Procedural documents:	*Umbilical Catheter Insertion and Management (Northern Services Only)*
Key Words:	*Umbilical Catheter* *Blood pressure Transducer* *Arterial Blood Sampling*
In consultation with and date:	
Contact for Review:	*SSN Alicia Rogers*

KEY POINTS OF THIS PROCEDURAL DOCUMENT:

- Identification of the need for arterial blood transducer monitoring
- Safe priming of blood transducer monitoring, management and blood sampling of line.

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1. INTRODUCTION

1.1. There are several methods of neonatal blood gas measurement which include:

- Indwelling arterial catheters, e.g. umbilical or peripheral arterial line (which also allow invasive blood pressure monitoring)
- Peripheral arterial 'stab' puncture sample
- Capillary blood sample (commonly taken from a heel prick)
- Non-invasive methods: oxygen saturation monitoring, end-tidal carbon dioxide (CO2) monitoring and transcutaneous oxygen tension/CO2 monitoring.

The most accurate and non-invasive method of measuring oxygenation is oxygen saturation monitoring. Indwelling arterial catheters are a practical, reliable and accurate method of measuring acid-base parameters, provided they are inserted and maintained with the proper care. Capillary blood gas sampling is accurate, and a good substitute for radial 'stab' arterial puncture, avoiding many of the complications of repeated arterial puncture.

2. PURPOSE

2.1. The Standard Operating Procedure (SOP) has been written to:

- Identify the safe procedure of priming the blood transducer for attachment to the umbilical arterial catheter to enable continuous blood pressure monitoring and access for regular blood tests, especially blood gases
- Identify the procedure for the safe withdrawal of blood via an arterial catheter
- Reduce the risks associated with arterial blood sampling

3. ABBREVIATIONS

- UAC - Umbilical Arterial Catheter
- UVC - Umbilical Venous Catheter

4. DUTIES AND RESPONSIBILITIES OF STAFF

4.1. Role of Paediatric Medical Team:

- The Paediatric Medical Team are responsible for acting in accordance with this SOP
- Ensuring that risks associated with arterial blood gas sampling are minimised.

4.2 Role of Neonatal Staff:

The Neonatal Staff nurses are responsible for:

- Assisting the doctor in the procedure
- Observing the baby for associated risks and complications and taking appropriate action
- On-going monitoring and care of arterial line

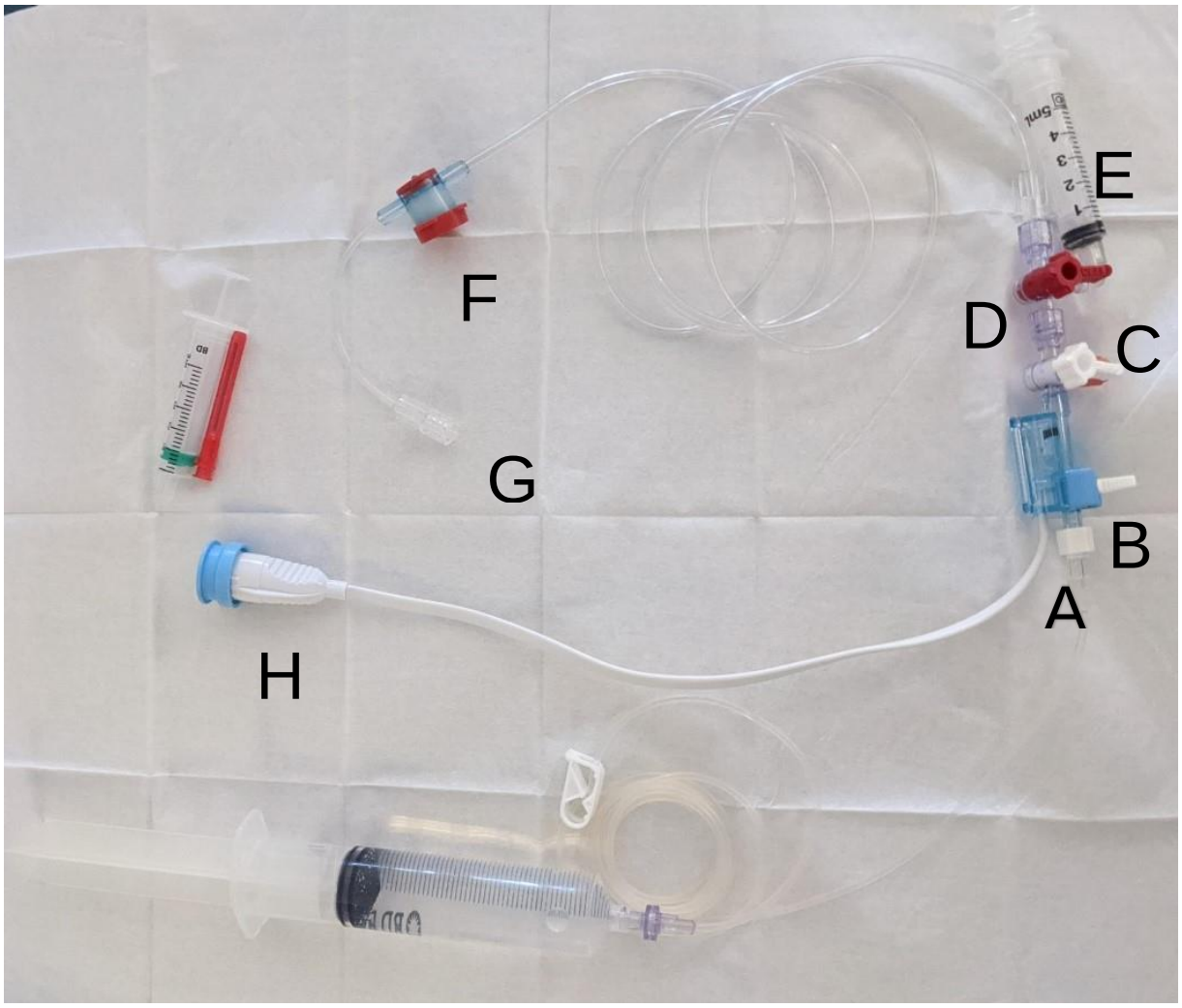
5. GUIDANCE TO SET-UP ARTERIAL BLOOD PRESSURE MONITORING (Maintenance fluids)

5.1 Use aseptic non-touch technique throughout the procedure.

5.2 Equipment Required:

- Prescription for fluids
- Braun syringe pump driver
- Braun giving set
- Pressure monitor transducer set
- 50ml and 1ml IV syringe
- Filter needles
- Heparin 1000u/ml or 50u hepsal
- NaCl 0.9% (normally) or 0.45% (in the extremely premature)
- Sterile gloves
- Chlorhexidine sanicloths for hub cleaning
- Yellow drugs added to infusion label

5.3 Diagram of arterial blood transducer set up



6. PROCEDURE FOR PRIMING BLOOD TRANSDUCER

1. Draw up 50 mls NaCl and 50u heparin and add to Na Cl making 1u heparin/ml Na Cl heparinised saline infusion flush.
2. Flush heparinised saline mixture through Braun giving set.
3. Take transducer set and connect to giving set **(A)**.
4. Now systematically flush heparinised saline flush through transducer. The solution will only advance when the blue transducer **(B)** is pinched together.
5. Close white tap **(C)** to baby end and open orange port.
6. Pinch blue plastic together and advance fluid until it drips out of orange port **(C)**.
7. Close orange port **(C)** and then close white tap **(C)** to orange port.
8. Close red tap **(D)** to baby end and flush (pinching blue plastic) heparinised saline into the enclosed syringe **(E)** 2.5mls.
9. Close red tap **(D)** towards transducer and holding the set upright with transducer at top. Ensure the bubbles/air at the top of the syringe.
10. Flush the heparinised saline 2.5mls from the syringe **(E)** through the line towards the baby end **(G).** Repeat step 9 & 10 until the line is primed and no air bubbles.
11. **ENSURE THE 2.5ML SYRINGE IS EMPTY POST PRIMING & FLUSHING.**
12. Label 50 ml syringe with contents.
13. The set is now flushed and ready to connect.
14. Place the 50ml syringe in the Braun infusion pump prior to attaching to the umbilical line to prevent siphonage. Purge the infusion using the pump facility to ensure there is no negative pressure within the infusion set prior to commencing.
15. Attach infusion to umbilical line.
16. Start infusion pump according to Dr's instructions (usually 0.5ml/hour).
17. Take blue cap off transducer pressure lead **(H)** and attach to lead from monitor.
18. Set monitor to display invasive arterial monitoring.
19. Place and always keep blue transducer in cot at level with infant's heart.
20. Zero pressure transducer.

If arterial monitoring does detect pulsatile pressures immediately then wait a few minutes or ask attending medical team to flush line or take sample.

7. ZERO AND CALIBRATE THE PRESSURE TRANSDUCER

Calibration ensures that the transducer pressure is zero prior to inline pressure measure.

7.1. Calibrate:

- Prior to monitoring commencing
- Following line sampling
- Following transducer disconnection
- If infant is moved
- If a poor trace

7.2 Process:

1. Ensure transducer is at level of infant's heart
2. Close tap to baby by orange hub next to transducer.
3. Open orange hub to room air
4. Press 'Zero' on the monitor menu bar and wait for instructions on screen to confirm transducer zeroed
5. Close orange hub to room air
6. Open tap by orange hub

Look for pulsatile wave on monitor to confirm reading

8. ARCHIVING ARRANGEMENTS

The original of this SOP will remain with the author SSN Alicia Rogers Special Care Baby Unit. An electronic copy will be maintained on the Trust intranet, G – Guidelines, P – Policies, SOP – Standard Operating Procedure. Archived electronic copies will be stored on the Trust's "archived policies" shared drive, and will be held indefinitely. A paper copy (where one exists) will be retained for 10 years, unless the document itself specifically-concerns children, in which case a paper copy will be retained for 25 years.

9. PROCESS FOR MONITORING COMPLIANCE WITH AND EFFECTIVENESS OF THE STANDARD OPERATING PROCEDURE

9.1. To evidence compliance with this policy, the following elements will be monitored:

What areas need to be monitored?	How will this be evidenced?	Where will this be reported and by whom?
Any concern or non-compliance with this guideline that is identified through the investigation of clinical incidents, complaints or claims will be reviewed as per Trust policy	audit and / or amendment to the guideline	*Neonatal Lead Consultant*

APPENDIX 1: COMMUNICATION PLAN

The following action plan will be enacted once the document has gone live.

Staff groups that need to have knowledge of the guideline/SOP	*Paediatric Medical Team* *Neonatal Staff*
The key changes if a revised document	*Nil – First Version of the SOP*
The key objectives	*This guideline sets out the process for the safe priming and management of arterial blood transducers and the procedure for taking arterial blood samples from a line*
How new staff will be made aware of the procedure/guideline and manager action	*Training and refresher in SCBU study days.* *Video of process of priming line and SOP in folder for easy access when required.*
Specific Issues to be raised with staff	*Neonatal Lead Consultant*
Training available to staff	*Training and refresher in SCBU study days.* *Video of process of priming line and SOP in folder for easy access when required.*
Any other requirements	*Nil*
Issues following Equality Impact Assessment (if any)	*Nil*
Location of hard / electronic copy of the document etc.	*Procedural document on the Trust intranet* *.*